

TCS349: Responsible and Explainable AI

Unit 3, Lecture 5

Emerging Skills & Workforce Transition in the AI Era

Which skills are rising in demand, the difference between reskilling and upskilling, and how workers can navigate a career path through AI-driven change.

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Emerging Skills



Reskilling & Upskilling



Workforce Transition



Career Mapping

1. Emerging Skill Requirements - What's Changing

As AI takes over routine tasks, demand shifts toward skills that complement rather than compete with automation.

Data literacy - the ability to interpret, question and work with data-driven outputs - is increasingly expected across many roles, not just technical ones.

Employers increasingly value the ability to work alongside AI tools effectively, even in roles that are not AI-specialist positions.

Skill requirements are changing faster than traditional education cycles, making continuous learning a practical necessity rather than an option.

2. Technical Skills for an AI-Augmented Workplace

Basic AI and data literacy: understanding what AI tools can do, their limitations, and how to interpret their outputs critically.

Prompting and tool direction: framing tasks clearly enough for AI systems to produce genuinely useful results.

Domain-specific AI application: knowing how AI tools apply within a specific field, such as finance, healthcare or engineering.

Technical depth still matters for those building and maintaining AI systems, but working with AI is now a broader expectation.

3. Durable Human Skills That Remain in Demand

Judgement and contextual decision-making: weighing factors an AI system cannot fully capture, especially in ambiguous situations.

Interpersonal and emotional skills: communication, negotiation and empathy remain difficult for AI to replicate convincingly.

Creative and original problem-solving: generating genuinely new approaches, rather than recombining existing patterns.

Ethical reasoning: recognising when a situation raises fairness, privacy or safety concerns that require human judgement.

4. Reskilling vs Upskilling - What's the Difference

Upskilling means deepening a worker's existing skill set to keep pace with how their current role is evolving.

Reskilling means training a worker for a substantially different role, often because their original role has been significantly automated.

Upskilling is typically less disruptive and can happen gradually alongside ongoing work.

Reskilling usually requires more structured support, since it often involves a genuine career change rather than a skill update.

5. Employer-Led Workforce Transition Strategies

Internal mobility programmes help displaced workers move into growing roles within the same organisation rather than leaving entirely.

Structured training partnerships with educational providers can deliver targeted skills faster than a worker retraining independently.

Phased transition plans give workers time to build new skills before their current role is significantly reduced.

Transparent communication about which roles are changing helps workers plan ahead rather than being caught by surprise.

6. Career Mapping - Planning a Path Through Change

Career mapping identifies a worker's current skills, a target role, and the specific skill gap between the two.

Adjacent-role mapping looks for roles that share a meaningful overlap with a worker's existing skills, reducing the retraining burden.

Mapping AI exposure alongside skill gaps helps prioritise which transitions are most urgent versus which can happen gradually.

Individual career mapping works best when paired with organisational data on which roles are actually growing or shrinking.

7. Barriers to Successful Workforce Transition

Time and financial cost: many workers cannot afford to pause earning income long enough to retrain substantially.

Access gaps: quality retraining programmes are not equally available across regions, industries or income levels.

Age and experience bias: older or more experienced workers sometimes face additional barriers when reskilling into new fields.

Without deliberate support, workforce transition can reinforce the inequality patterns discussed in the previous lecture rather than resolve them.